



Evolving Perceptions of AI Use and Academic Integrity: Insights from EFL Learners in Turkish Higher Education

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Abstract

Using a qualitative, multiple cross-sectional research design, this study investigated the evolving perspectives on the use of artificial intelligence (AI) and academic integrity between two cohorts of English as a Foreign Language learners ($n = 48$ and $n = 96$, respectively) enrolled in an English Language Preparatory Program at a Turkish university over two academic terms. The study revealed three significant patterns in students' changing perspectives: a shift from broad acceptance or rejection to detailed consideration of appropriate AI use, enhanced ethical awareness encompassing attribution and plagiarism, and recognition of AI as an integral educational tool that requires thoughtful management. These changes reflect a growing understanding of AI's role and implications in academic practice. The findings align with recent global research on AI integration in education, particularly regarding students' mixed perceptions of AI tools' potential benefits and ethical concerns. The study suggests that educational institutions develop context-specific guidelines and support systems that balance technological innovation with academic integrity.

Keywords Academic integrity · Artificial intelligence in education (AIED) · Academic coursework · EFL learners' perceptions · Turkish higher education · English language preparatory program

Introduction

The integration of Artificial Intelligence (AI) into education—particularly through technologies such as machine learning and natural language processing—has become increasingly prevalent, offering the potential to enhance and personalize learning experiences (Jiang, 2022; Rane et al., 2023). A rapidly advancing branch of AI, generative AI (GenAI),

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leverages machine learning algorithms, neural networks, and vast datasets to produce original content, including text, images, and audio (Ooi et al., 2023). This technological evolution has had a profound impact across a range of sectors, with education standing out as a domain where the influence is both significant and multifaceted (Holmes et al., 2019; Jiang, 2022).

Artificial Intelligence in Education (AIED) promises personalized and adaptive learning pathways but also introduces complex challenges (Rane et al., 2023; Tiwari, 2023). These challenges span technological, pedagogical, and societal dimensions. Among them, the digital divide exacerbates educational inequality, further influencing learners' access to opportunities and deepening socioeconomic disparities (Li, 2023). Within institutional settings, the integration of AI technologies is often hindered by gaps between technological capabilities and classroom practices, insufficient AI literacy among educators, and resistance or scepticism from both teachers and students (Chiu et al., 2023).

One of the most pressing concerns associated with the rise of GenAI in educational contexts is its potential threat to academic integrity, a term defined as “compliance with ethical and professional principles, standards, practices, and a consistent system of values that serves as guidance for making decisions and taking actions in education, research and scholarship” (Tauginienė et al., 2018). GenAI tools enable the production of human-like content that can facilitate academic misconduct, raising significant concerns about plagiarism and authenticity verification in higher education (Eke, 2023; Sullivan et al., 2023). The implications extend to scholarly publications, where verifying genuine authorship has become increasingly complex (Bozkurt, 2024).

In this context, the present study investigates how English as Foreign Language (EFL) learners in Turkish higher education perceive and navigate the intersection of AI tool usage and academic integrity in their academic coursework. By capturing learner perspectives across different timeframes, this study provides critical insights into the evolving attitudes toward AIED and contributes to a deeper understanding of the ethical and pedagogical implications of integrating AI in language learning environments.

Review of Literature on AI Integration and Academic Integrity in EFL and Higher Education

The increasing presence of AI, particularly GenAI, in higher education has sparked widespread interest among scholars, educators, and policymakers. As AI continues to reshape the ways students access, produce, and interact with academic content, its pedagogical benefits and ethical risks are becoming increasingly evident. In foreign language education, this dual impact is especially pronounced, where tools such as ChatGPT can facilitate personalized learning while simultaneously challenging traditional academic integrity frameworks.

A growing body of literature has highlighted the ethical challenges posed by GenAI in academic settings. Perkins (2023) observed that although GenAI systems enhance language acquisition through personalized feedback and tailored support, they also disrupt traditional assessment practices by enabling the production of seemingly original, AI-generated content. Similarly, Pan (2024) raised concerns about plagiarism and the authenticity of written work produced with GenAI assistance, particularly in EFL contexts. In a related study, Ofem et al. (2024) found that positive perceptions of ChatGPT among Nigerian university students were often linked to dishonest academic practices, reinforcing the need for robust institutional frameworks to guide ethical AI use.

In response to these concerns, scholars have emphasized the need for educational institutions to adopt clear, ethical, and inclusive policies for GenAI integration. Figueroa and Eaton (2023) highlighted the importance of balancing innovation with fairness and accessibility, urging institutions to establish clear guidelines. Eke (2023) similarly proposed that academic policies be revised to account for GenAI, including provisions for AI acknowledgment, educator training, and integrity policy updates. Bozkurt (2024) and Amigud et al. (2018) further stressed that fostering transparency, promoting AI literacy, and creating supportive environments are key to mitigating the risks associated with AI-enabled misconduct.

Despite these challenges, several studies have acknowledged the potential benefits of GenAI in enhancing student learning. For instance, Sevnarayan and Potter (2024) reported that students in a South African e-learning context valued GenAI for its ability to boost their confidence and academic performance. Likewise, Wiredu et al. (2024) found that a significant majority of students at three Ghanaian universities experienced improved learning outcomes with GenAI tools, while a similar proportion also voiced concerns about academic integrity. These findings underline the importance of adopting balanced approaches that harness AI's pedagogical value while safeguarding ethical academic practices.

Furthermore, studies across diverse cultural and disciplinary contexts have highlighted a range of learner perceptions regarding AI's role in academic integrity. For example, Gruenhagen et al. (2024) found that many Australian university students used GenAI tools for assessments without perceiving it as academic misconduct, suggesting a need for clearer institutional guidance. Similarly, Yusuf et al. (2024), in a cross-cultural investigation, observed high levels of GenAI awareness and usage, emphasizing the importance of culturally responsive policies to address ethical concerns. Additionally, in a Japanese higher education context, Price (2024) reported that EFL students appreciated the efficiency of ChatGPT in completing academic tasks but also expressed doubts about its reliability and ethical implications, thus reinforcing the need for human oversight in GenAI-supported learning. Likewise, Barrett and Pack (2023), in their study of American university students and educators, observed widespread uncertainty and a lack of preparedness in using GenAI for writing tasks, highlighting the urgency of developing explicit institutional guidelines and professional development. Complementing these findings, Busch and Bilgin (2014) argued that understanding student perspectives is essential for developing effective strategies to prevent academic misconduct and emphasized the value of student voice in shaping fair and responsive policies.

While these studies offer valuable insights into GenAI's impact on academic integrity in higher education, there remains a noticeable gap in research focused specifically on EFL contexts. Notable exceptions include the work of Barrett and Pack (2023) and Price (2024), yet comprehensive studies investigating the evolving perceptions of EFL learners regarding GenAI and academic integrity—especially over time—are still limited.

To address this gap, the present study aims to explore EFL learners' perceptions of AI tool usage in relation to academic integrity in their academic coursework within the context of Turkish higher education. By examining learner perspectives across two distinct cohorts over different time intervals, the study seeks to offer a longitudinal view of how these perceptions evolve. The investigation is guided by the following research questions:

1. How do EFL learners from two different cohorts, over distinct time intervals, perceive the use of AI tools in their academic coursework in relation to academic integrity?
2. How have EFL learners' perceptions of AI tools and academic integrity evolved over time?

Methodology

Research Context

The research was implemented in the English Language Preparatory Program within the School of Foreign Languages (SFL) at a state university in Türkiye. The program comprises two distinct programs based on students' majors: the General English Preparatory Program and the English Language Teaching/English Language and Literature Preparatory Program. This study focused on both programs, which offer intensive, yearlong general English and English for Academic Purposes language courses from A1 to C1 levels to students admitted to the various departments whose medium of education is English. The program provides comprehensive English courses that combine general English instruction with specialized skills training to help students develop the necessary English proficiency to succeed in their academic programs.

Instructors in the program are generally familiar with academic integrity and strive to uphold it through various efforts, such as using similarity detection tools and incorporating lessons on academic integrity principles into their courses. Additionally, students are required to complete academic tasks as part of the assessment system, including composing essays, preparing presentations, and writing reports. Mere observation suggests that many students use AI tools when completing these tasks, either for assistance or for practice.

While the plagiarism policy outlined in the course syllabi provides a clear definition of plagiarism—including the improper use of AI-generated content—and specifies penalties ranging from a zero grade to disciplinary action, there is no formal structure in place to track its implementation. Beyond the brief mention in the syllabi, there are no documented procedures to ensure consistent enforcement across courses nor mechanisms to systematically monitor or report plagiarism cases.

The policy itself adopts a strict, punitive approach to academic integrity, emphasizing deterrence through immediate penalties and escalating consequences for repeated violations. However, without an institutional framework for tracking its application, it remains unclear how consistently the policy is enforced or whether students receive uniform treatment across different courses and instructors. This lack of an implementation framework raises important concerns about how plagiarism cases are identified, recorded, and addressed beyond individual instructor discretion. Establishing a systematic tracking mechanism could enhance transparency, ensure consistency, and support a more educative approach to academic integrity.

Research Design

The study followed a qualitative, multiple cross-sectional research design. Given the rapidly changing landscape of AIED, a qualitative approach was considered suitable to explore EFL students' perceptions of AI and its place in academic coursework regarding academic integrity. Additionally, the multiple cross-sectional aspect provided the opportunity to compare and pin down the evolving perceptions across two cohorts, one at the end of the spring term of 2022–2023 and the other at the end of the fall term of 2023–2024. This comparison was significant, as the study period saw a rapid rise in AI tools and their integration into academic life, potentially shaping learners' perceptions, attitudes, and behaviors. The research design is visually presented in Fig. 1.

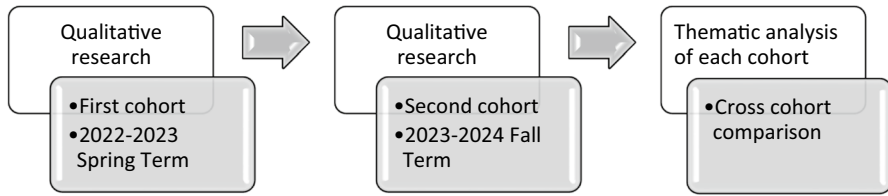


Fig. 1 Research design of the study

Participants

Two different cohorts of students at two distinct points in time studying EFL in the SFL at a Turkish state university took part in this study. In the 2022–2023 spring term, approximately 500 students were enrolled, and this number increased to around 600 in the 2023–2024 fall term. During the first data collection phase, 60 students responded to the online survey, while 113 students participated in the second phase. Of these, 48 students from the first phase and 96 from the second reported using AI for learning and practicing English and completing academic tasks. Hence, the analysis and reporting of findings were based on these responses. A convenience sampling method with criteria including active enrollment in the program and voluntary consent was employed. Participant demographics, including age, gender, and major, are displayed in Table 1.

As Table 1 shows, 68.75% of the first cohort were female, and the majority were aged 18–20 (70.83%). The second cohort had a slightly higher proportion of females at 71.88%, with the majority also in the 18–20 age range (93.75%). They were predominantly from English Language Teaching and English Language and Literature BA programs and other majors such as Management, Biology, and Economics.

Ethical Considerations and Approval Process

This study was approved by the university's Ethical Review Board, approval number E-84026528-050.01.04-2300111702 for the first and E-84026528-050.01.04-2300319771 for the second data collection phase. The participants were informed about the purpose of the study, and informed consent was obtained before participation in both data collection phases. Furthermore, measures were taken to ensure the confidentiality and anonymity of the participants, including the removal of personal identifiers from the data, the use of coded identifiers instead of real names, and adherence to ethical guidelines approved by the relevant institutional review board.

Data Collection Tool

A survey questionnaire was developed based on existing literature on AI tools, academic integrity, and language learning. The questionnaire, reviewed by two experts in the field to ensure content validity, included closed-ended questions to gather quantitative data about participant characteristics and open-ended questions to collect qualitative data about participant insights related to academic integrity for using AI for academic coursework. Data were collected online through Google Forms over two academic terms, at the end of the

Table 1 Participant characteristics

Characteristic	1 st Cohort		2nd Cohort	
	<i>n</i>	%	<i>n</i>	%
Gender				
Female	33	68.75	69	71.88
Male	15	31.25	26	27.08
Unspecified	0	0	1	1.04
Age				
18–20	34	70.83	90	93.75
21–23	11	22.92	5	5.21
24–27	3	6.25	1	1.04
Major				
English Language	10	20.83	28	29.17
Teaching	11	22.92	29	30.21
English Language	8	16.67	11	11.46
and Literature	3	6.25	2	2.08
Management	4	8.33	2	2.08
International Rela-	4	8.33	9	9.38
tions	2	4.17	2	2.08
Biology	1	2.08	0	0
Economics	1	2.08	1	1.04
Computer Engineer-	1	2.08	10	10.42
ing	1	2.08	0	0
International Trade	1	2.08	1	1.04
and Management	1	2.08	1	1.04
Environmental Engi-				
neering				
Molecular Biology				
and Genetics				
Archaeology				
Bioengineering				
History				
Total	48	100	96	100

spring term of 2022–2023 and the fall term of 2023–2024. The participants were invited to complete the survey via in-class announcements. To increase the response rate, reminders were sent, and the importance of the study was emphasized.

Data Analysis

To analyze the responses to the open-ended questions, we first translated the original responses from Turkish into English before we implemented a thematic analysis. Thematic analysis is a method for identifying, analyzing, and reporting patterns (themes) within qualitative data, focusing on participants' perceptions and meanings (Saldaña, 2013). The thematic analysis process was iterative and involved multiple stages of familiarization, coding, and theme development. The coding process was conducted in several rounds to ensure thorough immersion in the data and identification of key themes. This iterative process ensured that the final themes reflected the richness and diversity of participant perspectives. The researchers followed a collaborative approach, where the first researcher led the analysis, focusing on coding and identifying key themes while the second researcher validated the findings, suggesting additional themes, revisions, and alternative interpretations.

Disagreements were discussed and resolved through consensus. This approach ensured the rigor of the analysis and the accuracy and completeness of the findings. Figure 2 presents the stages involved in thematic analysis as described by Saldaña (2013):

Findings

Research Question 1: How do EFL Learners from Two Different Cohorts, Over Distinct Time Intervals, Perceive the Use of AI Tools in their Academic Coursework in Relation to Academic Integrity?

First Cohort

The thematic analysis regarding the EFL students' perceptions of AI use in their academic coursework and academic integrity revealed four main themes: Conditional Acceptance, Ethical Concerns, Learning Benefits, and Neutral Stance. These themes reflect the students' complex and nuanced perspectives about the intersection of AI tools and academic integrity in language learning and academic coursework (see Table 2).

Theme 1: Conditional Acceptance: Boundaries and Limitations of AI Use The most prevalent theme, Conditional Acceptance, received 20 mentions across two subthemes. It reflects students' openness to using AI tools within defined boundaries, acknowledging their potential value while emphasizing the need for clear limitations.

The first subtheme, *Acceptable as a Supporting Tool*, the most common subtheme with 12 mentions, indicated that students viewed AI tools as legitimate aids when used for specific supporting functions. One participant noted, "I think it can be seen as ethical since we don't have another source to check our homework" (R#19). This perspective was particularly evident in contexts where AI was used for learning purposes, as stated by one student, "I consider it ethically correct when an AI is used efficiently and in an instructive way." (R#7). Another student elaborated on this view, stating, "I do not see an ethical problem if the assignment is not completed entirely by using AI. I think it should be used as a preparation for the homework part and left at that stage" (R#30).

The second subtheme, *Moderation in Use* (8 mentions), highlighted students' emphasis on maintaining a balance between AI assistance and personal effort. This was clearly articulated by one participant who stated, "I strongly disapprove of zero effort on the part of the person; in such cases, help can be sought, but it must be in moderation and individual effort must be made" (R#41). In another response, the partial use of AI in assignments was regarded as acceptable: "We can improve our homework with a little help, but for the most

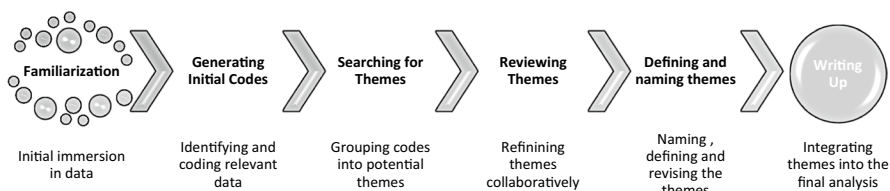


Fig. 2 Saldaña's framework of the thematic analysis process

Table 2 Perceptions of first cohort students regarding AI use in academic coursework and academic integrity

Main Theme	Subtheme	Supporting Tool	Frequency	Representative Quotes (Response #)
Conditional Acceptance: Boundaries and Limitations of AI Use	Acceptable as a Supporting Tool		12	"I find it right to use AI in homework if it is sufficient, if the meaning is unknown, and if the sentence structure is examined." (R#39)
	Moderation in Use		8	"I strongly disapprove of zero effort on the part of the person; in such cases, help can be sought, but it must be in moderation and individual effort must be made." (R#41)
	Plagiarism and Cheating		7	"It can create problems with honesty as it can create situations such as cheating and plagiarism." (R#3)
Ethical Concerns: Balancing Innovation with Academic Integrity	Hindrance to Learning		9	"I find it unethical. The assignments are given to us to improve ourselves. If we use these tools frequently to do our homework and make them a habit, we cannot improve ourselves." (R#24)
	Learning Enhancement		6	"The purpose of AI is to make our lives easier and save us time. While not entirely, I see no harm in using AI occasionally for assignments. After all, the goal is to learn." (R#21)
Learning Benefits: AI as an Educational Tool	Practical Utility		5	"Using AI tools gives us so many options for language learning. I don't think it has much negative impact on academic integrity." (R#5)
	Context-Dependent Views		4	"I think it is neither ethical nor unethical, so I have no comment." (R#29)
Neutral Stance				

part, it would be wrong to use AI"(R#14). This subtheme suggests that students recognize the need for self-regulation and are mindful of its limitations in their use of AI tools.

Theme 2: Ethical Concerns: Balancing Innovation with Academic Integrity The second major theme, Ethical Concerns, emerged with 16 total mentions, highlighting students' awareness of potential ethical challenges associated with AI use. This theme was divided into two subthemes: *Plagiarism and Cheating* (7 mentions) and *Hindrance to Learning* (9 mentions).

Concerns about plagiarism and academic dishonesty were explicitly addressed by several participants, with one stating, "It can create problems with honesty as it can create situations such as cheating and plagiarism"(R#3). The higher frequency of mentions regarding learning impediments (9 mentions) suggests that students were particularly concerned about the potential negative impact on their educational development. This was powerfully expressed by one participant who mentioned the lack of original contribution when AI is used in writing assignments by stating, "I don't think it is very ethical because most of the students use such programs in their writings, etc., and do not produce anything themselves"(R#37).

Theme 3: Learning Benefits: AI as an Educational Tool The third theme, Learning Benefits (11 mentions), revealed students' recognition of AI's potential positive contributions to their learning process. Within this theme, *Learning Enhancement* (6 mentions) emerged as slightly more prominent than *Practical Utility* (5 mentions). Students who identified learning benefits often emphasized the tool's role in facilitating the learning process, as exemplified by one participant who stated, "I find it right because it is also a learning tool"(#39).

Theme 4: Neutral Stance The fourth theme, Neutral Stance (4 mentions), reflects participants who maintained a neutral position on AI use, neither strongly advocating for nor opposing it. The primary subtheme, *Context-Dependent Views*, emerged from four participant responses that suggested their acceptance of AI varied based on the specific situation. This measured stance is exemplified by R#13, who described their position as "medium," suggesting that their judgment depends on the specific context in which the AI tool is used. This stance demonstrates participants' reluctance to make broad generalizations about AI ethics, preferring instead to withhold definitive judgment. Additionally, this neutral stance may stem from the complexity of the issue or a lack of sufficient information, as the initial data collection occurred when AI tools were relatively new to the participants. It may also be influenced by a fear of being identified as an AI user—despite assurances of privacy, some students may still worry that their teachers could discover their AI use and impose penalties. In contrast, responses from later data collection tend to be more definitive, suggesting a possible shift in understanding or familiarity with the topic or greater recognition of AI tools utilized for learning and carrying out academic tasks.

In sum, a notable finding across these themes was the evident tension between viewing AI as a helpful learning tool and maintaining academic integrity. This tension was particularly visible in the interplay between the Conditional Acceptance and Ethical Concerns themes. While students recognized AI's potential benefits, they also expressed clear reservations about its unrestricted use. This tension in students' perspectives highlights the complex relationship between viewing AI as a learning support tool and maintaining academic autonomy.

Second Cohort

The analysis of the second dataset, comprising 96 responses, revealed a more nuanced and contextualized understanding of AI use in academic settings. Four primary themes emerged from this analysis: Contextual Appropriateness, Plagiarism and Attribution, Learning Impact, and Technological Reality. These themes reflect a maturing discourse around AI use in academic contexts, with students demonstrating increased awareness of specific conditions and considerations for ethical AI integration (see Table 3).

Theme 1: Contextual Appropriateness: Defining Boundaries of Acceptable Use The most prominent theme in the second dataset was Contextual Appropriateness, with 42 total mentions across two subthemes. This theme reflected students' sophisticated understanding of when and how AI tools should be employed in academic work.

The first subtheme, *Purpose-Specific Usage* (23 mentions), demonstrated students' clear differentiation between acceptable and unacceptable purposes for AI use. As one participant articulated, "There is nothing unethical about getting ideas, but fully relying on them is unethical" (R#25). This distinction between AI as a verification tool versus a replacement for original work was frequently emphasized. Another student elaborated on this perspective, noting, "If the person has no idea about the subject, they can use it to get an idea, but using AI just because they are too lazy to do it themselves is not good" (R#70).

The second subtheme, *Degree of Usage* (19 mentions), highlighted students' awareness of the importance of moderation in AI use. This was explicitly quantified by one participant who stated, "The weight of AI use in homework should not exceed 5% of the total work" (R#61), suggesting a precise consideration of appropriate limits. This attention to quantifiable boundaries represents a more sophisticated approach to managing AI integration in academic work.

Theme 2: Plagiarism and Attribution: Enhanced Focus on Academic Integrity The second major theme revealed an increased emphasis on plagiarism concerns and proper attribution practices (25 total mentions). This theme demonstrated students' growing awareness of the relationship between AI use and academic integrity standards.

In the *Plagiarism Concerns* subtheme (17 mentions), students expressed clear views about the boundaries between assistance and academic dishonesty. One participant stated, "I don't think there is a problem with using it as long as it's done in moderation, but using it to plagiarize without putting in any effort ourselves is not ethically appropriate" (R#30). This sentiment was echoed across multiple responses, indicating a collective understanding of the ethical implications of AI use.

The *Attribution Requirements* subtheme (8 mentions) represented a new dimension in students' thinking about AI use, with explicit attention to sourcing and acknowledgment. As one student noted, "There is no problem as long as credit is given" (R#27). This attention to attribution suggests an evolving understanding of how AI tools should be integrated into academic citation practices.

Theme 3: Learning Impact: Balancing Educational Benefits and Risks The third theme focused on the educational implications of AI use, with students demonstrating awareness of both potential benefits and risks to their learning process. This theme encompassed 22 total mentions across two subthemes.

Table 3 Perceptions of second cohort students regarding AI use in academic coursework and academic integrity

Theme	Subtheme	Frequency	Representative Quotes (Response #)
Contextual Appropriateness: Defining Boundaries of Acceptable Use	Purpose-Specific Usage	23	"I prefer it to be used for checking rather than completing assignments" (R#13)
	Degree of Usage	19	"Its use in the entirety of homework is not ethical, but it can be used in some parts" (R#52)
Plagiarism and Attribution: Enhanced Focus on Academic Integrity	Plagiarism Concerns	17	"It can be used for assistance where necessary, but intellectual theft is not ethical" (R#2)
	Attribution Requirements	8	"Those who use it should state it as a source" (R#64)
Learning Impact: Balancing Educational Benefits and Risks	Self-Development Concerns	12	"Although it might seem good for the student, it negatively affects the student's development" (R#50)
	Learning Benefits	10	"The important thing is our improvement. If we correctly address the parts we are struggling with and move forward, I don't think there will be a problem" (R#48)
Technological Reality: Pragmatic Acceptance of AI in Education	Technological necessity	7	"The boundaries of technology and AI are expanding. We need to keep up with this trend and benefit from it" (R#87)

The *Self-Development Concerns* subtheme (12 mentions) reflected students' awareness of potential negative impacts on their learning experience. One participant observed, "Homework exists for you to practice. Doing it with the help of AI would be deceiving yourself, and I think it is not beneficial for your development" (R#44). This concern was balanced against the *Learning Benefits* subtheme (10 mentions), where students recognized AI's potential as a learning aid, with one noting, "I think it's necessary because if we don't have a strong educational background or sufficient knowledge, we can use these tools to check our homework, and I believe they contribute a lot to our learning" (R#46).

Theme 4: Technological Reality: Pragmatic Acceptance of AIED A notable new theme emerging in the second dataset was the recognition of AI as an inevitable part of modern education (7 mentions). This theme reflected a mature understanding of the changing educational landscape. As one student articulated, "I think it (using AI) can be permitted from time to time. After all, we live in a technological age. These aids are becoming normalized because we are evolving day by day" (R#40). This perspective suggests a shift from viewing AI use as a choice to seeing it as a necessity requiring careful management.

Research Question 2: How have EFL Learners' Perceptions of AI Tools and Academic Integrity Evolved Over Time?

Figure 3 visually represents the identified themes and subthemes from the two cohorts, illustrating their perceptions of AI usage and academic integrity over time.

For the first cohort, themes such as Conditional Acceptance and Ethical Concerns dominate, revealing a relatively cautious and exploratory attitude towards AI. In contrast, the second cohort's responses reflect a shift towards Contextual Appropriateness and Technological Reality, suggesting a more nuanced understanding of AI use in academic contexts. This cohort demonstrated greater awareness of context-specific applications, practical limitations, the necessity of attribution, and a growing recognition of AI as an integral part of modern education. Therefore, it is evident that a significant shift occurred in how EFL preparatory program students perceive and conceptualize AI use in academic contexts. In sum, building on these findings, it is possible to synthesize a broader understanding of students' perceptions of AI use and academic integrity, which can be categorized into three major shifts: increasing sophistication in ethical reasoning, enhanced contextual awareness, and growing recognition of AI as a technological reality (see Fig. 4).

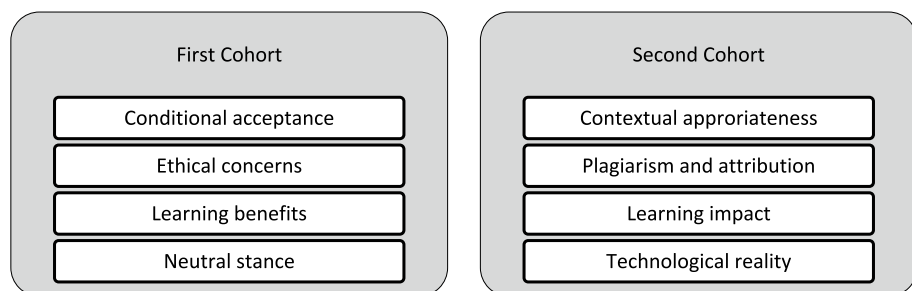


Fig. 3 Identified themes and subthemes from both student cohorts

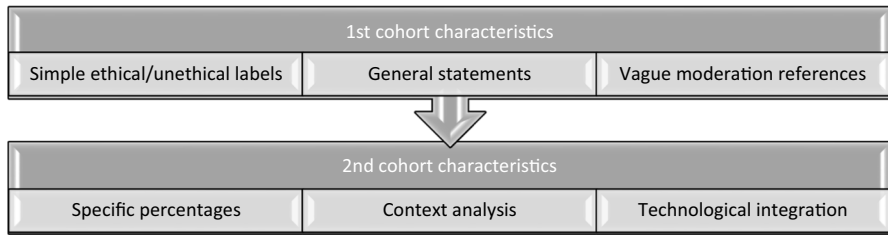


Fig. 4 Major shifts in EFL students' perceptions of AI use and academic integrity across two cohorts

Evolution of Ethical Understanding The most notable development between cohorts was the transition from binary ethical positions to more nuanced perspectives on AI use. While the first cohort typically expressed categorical acceptance or rejection of AI tools through simple "ethical" versus "not ethical" statements, the second cohort demonstrated more sophisticated ethical reasoning that considered contextual factors and specific use conditions. This maturation is particularly evident in the emergence of quantified guidelines, such as suggestions that "AI use in homework should not exceed 5% of the total work," (R#61), indicating a move toward more practical and nuanced ethical frameworks.

Enhanced Contextual Awareness The second cohort exhibited a markedly more sophisticated understanding of context-appropriate AI use. Unlike the students in the first cohort, who focused primarily on immediate classroom implications, these students demonstrated a broader awareness of academic and professional contexts. This expanded perspective was accompanied by more detailed considerations of specific use cases, such as distinguishing between using AI for grammar checking versus complete assignment completion. The evolution suggests a growing capability among students to differentiate between appropriate and inappropriate AI applications across various academic scenarios.

Technological Integration Perspectives A striking difference between cohorts was the emergence of more pragmatic views regarding AI's role in education. While the first cohort predominantly focused on immediate learning impacts, the second cohort recognized AI as an inevitable educational tool, exemplified by statements acknowledging that "as long as there is the internet, it will be a system that will be with us in every aspect of our lives" (R#87). This shift from reactive to proactive thinking about AI integration suggests a maturing understanding of technology's role in academic environments.

This analysis also reveals a maturation process that can be summarized through three key stages: from initial binary views of AI usage to a more nuanced understanding and finally towards the practical integration of AI in academic work (see Fig. 5).

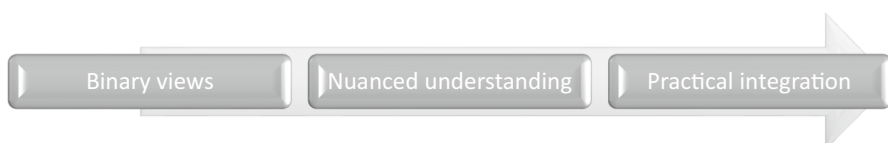


Fig. 5 Stages of contextual and ethical maturation in AI use

Discussion and Conclusion

This study explored the perceptions of AI use and academic integrity among EFL learners across two cohorts at two different time intervals, examining potential changes in their understanding. The findings reveal evolving perceptions of AI use and academic integrity among EFL learners across two cohorts, displaying a significant progression in their understanding and attitudes over time. The thematic analysis emphasized nuanced perspectives on the opportunities and challenges associated with integrating AI tools in academic contexts.

The first cohort primarily exhibited conditional acceptance, acknowledging AI's value as a supportive tool while emphasizing the need for clear boundaries to preserve academic integrity. Their concerns centered around ethical challenges, such as plagiarism and the potential for AI to hinder genuine learning, although they also recognized its educational benefits. A neutral stance, expressed by a minority, reflected uncertainty or a context-dependent view of ethical dilemmas, which may be explained by the fact that during the first phase of data collection, GenAI tools had only recently been introduced; consequently, there was a low level of recognition and limited use of these tools at the time.

In contrast, the second cohort displayed greater contextual awareness and pragmatic acceptance of AI's role in education. Themes such as contextual appropriateness and technological reality emerged, reflecting a more sophisticated discourse. This group advocated for explicit guidelines and demonstrated a deeper understanding of AI as an integral part of modern education, marking a transition from binary views toward a balanced integration of AI into academic practices.

The findings of this study align with and diverge from previous research in certain ways, highlighting both similarities and distinct contributions. Consistent with existing literature, the study underscores the dual-edged nature of GenAI tools, reflecting both their educational potential and ethical challenges. Similar to Price (2024), who reported Japanese EFL students' recognition of GenAI's efficiency alongside ethical concerns, this study also found that students were aware of both the advantages and potential risks of using AI in academic contexts. Additionally, the results echo Barrett and Pack's (2023) findings that students and educators often lack clear guidelines for ethical AI use, suggesting the need for comprehensive institutional policies.

However, this study diverges from some prior works by capturing a distinct evolution in learners' perceptions over time. Unlike the snapshot perspectives provided by some researchers such as Gruenhagen et al. (2024) and Ofem et al. (2024), this study with a multiple cross-sectional design examines two cohorts at different time intervals, revealing a shift from initial caution to a more nuanced and contextually informed understanding of AI use. For instance, while the earlier cohort emphasized concerns such as plagiarism and over-reliance on AI, the later cohort highlighted themes like contextual appropriateness and the necessity for human oversight as reported by Price (2024) as well, reflecting growing familiarity and critical engagement with AI tools.

The distinct nature of this study lies in its comparative approach regarding time interval, which examines how perceptions of AI and academic integrity evolve over time within the same educational context. To date, no other research has specifically investigated two distinct student cohorts' perspectives on AI use in academic work and its implications for academic integrity in a multiple cross-sectional manner. This temporal and comparative focus allows for a deeper understanding of how learners adapt to and

internalize the implications of emerging technologies, setting this research apart from prior studies that predominantly analyze single cohorts or single cross-sectional data.

Furthermore, by situating its investigation within the EFL context, this study addresses a significant gap noted in the introduction, where limited research exists on the intersection of AI and academic integrity in language learning environments. This highlights the specific challenges and opportunities AI poses for EFL learners, particularly in fostering critical awareness and ensuring ethical usage of AI tools in the language learning process and academic coursework.

In conclusion, while this study corroborates many findings in existing literature regarding AI's benefits and challenges, its comparative design spanning distinct timeframes offers a novel lens for understanding the evolving perceptions of AI in FLE. The findings highlight a clear progression among Turkish EFL learners, transitioning from foundational ethical concerns to a more balanced and informed understanding of AI's role in education. This maturation, characterized by enhanced ethical reasoning and contextual awareness, parallels global trends observed in the literature, emphasizing the importance of institutional policies that align with students' evolving perspectives. By addressing these shifts, higher education institutions can develop responsive and forward-thinking policies and pedagogical strategies that balance AI's transformative potential with the preservation of academic integrity. This approach supports the integration of AI as an indispensable element of modern education while fostering an educational environment that embraces technological advancements responsibly and ethically.

Limitations of the Study and Suggestions for Future Research

While this study offers unique contributions, it is not without limitations. First, the sample size, though adequate for qualitative analysis, and the study's focus on a Turkish higher education context limit the generalizability of the findings to other cultural, educational, or linguistic contexts. Additionally, although the study uses thematic analysis of data gathered through a written survey questionnaire, deeper qualitative insights could be obtained through more in-depth methods like interviews.

These limitations suggest areas for future research, such as expanding the study to different contexts or employing mixed methods to gain a more comprehensive understanding. Moreover, future researchers can also investigate the effectiveness of training interventions designed to promote ethical AI use and examine whether such programs improve students' understanding and application of academic integrity principles. Finally, research should explore pedagogical adaptations, including effective methods for teaching AI literacy and ethical use. Such investigations would provide valuable insights for institutions seeking to optimize AI integration while preserving educational quality.

Implications

The findings of this study offer valuable implications for higher education institutions, educators, and policymakers in navigating the integration of AI, particularly in EFL contexts. By drawing on insights from the literature and the research findings, several key recommendations emerge to ensure the ethical and effective use of AI tools in academic environments.

First, comprehensive institutional frameworks are essential to balance the benefits and risks of AI usage. Prior research (e.g., Barrett & Pack, 2023; Figueroa & Eaton, 2023)

underscores the need for clear guidelines to address concerns surrounding academic integrity and mitigate misuse. This study highlights the gap in students' understanding of acceptable AI practices, pointing to the urgency of policies that provide frameworks and practical examples, address ethical dilemmas, and promote transparent attribution of AI-generated work. Institutions must ensure these policies remain flexible, adapting to evolving technological capabilities while maintaining rigorous academic standards.

Second, this research emphasizes the importance of ongoing education and AI literacy programs. As revealed by this study, students' familiarity with AI tools significantly influences their ethical considerations over time. Institutions should invest in training programs to equip learners with the skills to critically evaluate and responsibly utilize AI. These programs should address issues such as plagiarism, over-reliance, and the ethical implications of AI in academic work, aligning with recommendations by Sullivan et al. (2023) and Mahendra et al. (2023). It is also essential to organize and provide ongoing professional development opportunities for EFL teachers to effectively integrate GenAI tools into education while upholding principles of academic integrity, as suggested by Barrett and Pack (2023).

Third, the study underscores the need for collaborative approaches among educators, policymakers, and technology developers to co-create ethical norms for AI use as suggested by Pan (2024). Instructors should facilitate dialogues that address student concerns, explore AI's potential to enhance learning, and emphasize the importance of human oversight. For EFL contexts, educators should design tasks that balance AI-assisted learning with opportunities for authentic language production. This involves moving beyond simple prohibitive measures to embrace nuanced frameworks that recognize AI as an indispensable feature of modern education. Policies should promote the integration of AI into academic integrity frameworks, focusing on fostering self-regulation, technological literacy, and ethical decision-making. Policymakers should create a culture of academic integrity by establishing clear standards, guidelines, and procedures for all stakeholders (Razi et al., 2022), complemented by fostering this culture through initiatives like appointing academic integrity champions, providing education, implementing robust policy evaluations, and shifting the focus from misconduct to promoting integrity (Bretag & Mahmud, 2016).

Furthermore, this study's comparative design spanning distinct timeframes offers unique insights into how perceptions of AI and academic integrity evolve. Regularly gathering and analyzing learner feedback on AI integration can enable institutions to adapt their strategies to meet students' changing needs and technological advancements. This iterative approach ensures that policies remain relevant and effective in addressing emerging challenges.

To conclude, this study contributes to the discourse on AIED by illustrating the dynamic interplay between students' perceptions of AI, academic integrity, and institutional policies. Recognizing AI's transformative potential and its indispensable role in contemporary education, higher education institutions must implement evidence-based strategies informed by this and similar studies. By doing so, institutions can harness AI's benefits, uphold rigorous academic integrity, and foster ethical academic practices in EFL and beyond.

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Code Availability N/A.

Data Availability The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Consent to Participate N/A.

Consent for Publication N/A.

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